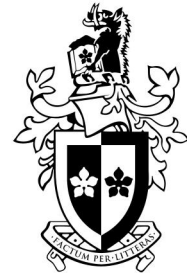


School of Science, Computing and Engineering Technologies
Software Architecture and Application
Assignment 1



Smart Fleet Management (SmartFM) Software Requirements Specification

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Abstraction

This Software Requirements Specification (SRS) document details the architectural foundation and system requirements for the Smart Fleet Management System (SmartFM), designed for ABC-Trans, an important logistics and transportation provider in Vietnam. Operating on fragmented, manual, and branch-specific software, ABC-Trans faces critical bottlenecks in vehicle allocation, real-time shipment tracking, and merged financial billing that limit enterprise scalability. To solve these operational challenges, the proposed SmartFM platform mixes logistics pipelines across all regional branches through a high-performance web and mobile ecosystem.

Applying "Tasks and Support" methodology, this specification defines the system boundaries, human-focused workflows, and technical constraints required to support high-concurrency operations. Key architectural patterns are integrated to safeguard transactional integrity and system performance: Command Query Responsibility Segregation (CQRS) and Read Replicas are used to separate heavy real-time telematics lookups from core state changes; Pessimistic Row-Level Locking is implemented within the scheduling engine to prevent resource conflicts and duplicate fleet bookings; and ACID-applied transaction strictly controls the automated bill pipeline to ensure unchangeable financial accuracy. Finally, this document serves as the blueprint for development, matching stakeholder operational goals with a scalable, data-driven software architecture ready for enterprise deployment.

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Part A: Introduction

The Software Requirements Specification (SRS) emphasizes the comprehensive functional and non-functional requirements for the Smart Fleet Management System (SmartFM) designed for ABC-Trans, one of Vietnam's most prominent logistics and transportation enterprises. Notably, ABC-Trans operates a nationwide network providing vital supply chain services to major department stores and supermarkets. In addition, the centralized ecosystem unifies enterprise-wide fleet tracking, driver allocations, booking pipelines, and financial transactions which overcomes operational scaling limitations caused by legacy, disconnected, branch-specific software and manual record-keeping.

Consequently, the main purpose of this report is to map out the system boundaries and business processes using the Tasks and Support approach. By focusing heavily on domain-level interactions and user workflows rather than pre-emptive database architecture, this specification ensures that the developed software natively resolves the technical challenges of high-volume, concurrent read operations including real-time tracking and high-integrity write transactions across all regional branches.

Part B: Project Overview

The Software Requirements Specification (SRS) emphasizes the comprehensive functional and non-functional requirements for the Smart Fleet Management System (SmartFM) designed for ABC-Trans, one of Vietnam's most prominent logistics and transportation enterprises. Notably, ABC-Trans operates a nationwide network providing vital supply chain services to major department stores and supermarkets. In addition, the centralized ecosystem unifies enterprise-wide fleet tracking, driver allocations, booking pipelines, and financial transactions, effectively resolving operational scaling limitations caused by legacy, disconnected, branch-specific software and manual record-keeping.

2.1 Domain Vocabulary

The following glossary defines the specialized business and technical terms foundational to the ABC-Trans operations and the SmartFM platform:

- SmartFM: A centralized system integrating tracking, operations, and fleet management across all ABC-Trans branches.
- Shipment Offering: A defined transportation service option categorized by vehicle capacity, routes, and rates.
- B2B Corporate Order: A high-volume logistics request placed by enterprise clients for bulk cargo movement.
- Vehicle Assignment: The formal allocation of an available fleet vehicle to an active shipment order.
- Driver Management: Operational workflows dedicated to monitoring operator availability, active service hours, regulatory compliance, and safety credentials.
- Real-Time Shipment Tracking: Live telemetry streams integrated with digital mapping interfaces that enable dispatchers and customers to monitor current cargo coordinates and shifting ETAs instantly.

- Invoice: A digital financial statement compiled by the system detailing services rendered and costs.
- Receipt: An unalterable electronic document provided to clients as immediate proof of a completed payment.
- Read Replica: Dedicated database instances used to offload high-frequency tracking and search queries.
- Pessimistic Row-Level Locking: A database control that locks data rows during booking to prevent truck double-bookings.
- CQRS: An architectural pattern separating data writes from data lookups to maximize performance.
- ACID Compliance: Database properties ensuring multi-step workflows either succeed completely or roll back entirely.

2.2 Project Goals

The primary purpose of the Smart Fleet Management System (SmartFM) is to transition ABC-Trans from fragmented, manual, and branch-specific legacy workflows into a unified, high-performance digital ecosystem. Specifically, the strategic goals and their corresponding operational incentives are outlined below:

- Centralize Enterprise Operations: Unify data silos across all regional Vietnamese branches into a single authoritative platform, ensuring that vehicle availability, driver compliance logs, and order queues are synchronized in real time.
- Maximize System Performance and Read-Write Throughput: Address structural bottlenecks during peak operational hours by decoupling heavy read actions (such as high-frequency GPS coordinate map refreshes by customers) from high-integrity write actions (such as B2B order creation), utilizing advanced architectural patterns like CQRS and Read Replicas.
- Eliminate Resource Scheduling Conflicts: Prevent operational errors such as the double-booking of logistics assets or drivers by enforcing strict transactional integrity via Pessimistic Row-Level Locking during the vehicle assignment workflow.
- Enhance Customer Transparency and Retention: Build trust with major enterprise clients, including supermarkets and department stores, by introducing reliable, sub-second-latency real-time shipment tracking and accurate, automated Estimated Time of Arrival (ETA) projections.
- Streamline Corporate Financial Workflows: Shorten payment cycles and reduce administrative errors by offering multi-method payment pathways such as cash, credit cards, and debit cards processed via database transactions that strictly adhere to ACID properties, ensuring unalterable automated invoices and receipts.
- Drive Data-Driven Resource Utilization: Provide executive leadership with high-fidelity historical reporting tools to evaluate driver service hours and fleet asset utilization across day-, week-, month-, and year-to-date intervals.

2.3 Assumptions

- **UI/UX Compliance:** Front-end developments will strictly follow ABC-Trans enterprise digital UI standard asset sheets (Doc Ref: ABC-UI-GUIDE-2026), eliminating the need for explicit wireframes in this architectural document.
- **Infrastructure Provision:** ABC-Trans will fully procure and provision all necessary mobile tablets, terminal nodes, and target cloud hosting environments with pre-configured runtime compatibility.
- **Network Stability:** Cellular data coverage is assumed to remain continuous and stable across all domestic transport corridors to support real-time GPS tracking transmission.
- **Academic Integrity:** All project deliverables are divided equally among group members and authored entirely through human collaboration without reliance on generative AI engines.

2.4 Project Scope

The proposed software ecosystem is engineered around a comprehensive suite of backend services and interfaces, specifically optimized to streamline fleet logistics, automate dispatching workflows, and elevate end-to-end customer service delivery.

2.4.1 In Scope

- **Secure Customer Web Portals:** Architecture and deployment of a secure web portal enabling clients to authenticate, manage corporate profiles, place cargo orders, and view real-time shipping manifests.
- **Mobile Cargo Tracking Interfaces:** Design of cross-platform mobile interfaces engineered to process and render continuous telemetry data, allowing users to track consignment locations and estimated times of arrival (ETA).
- **Dispatcher Assignment Dashboards:** Implementation of a centralized, high-concurrency dashboard with role-based access for dispatchers to dynamically allocate fleet resources, assign drivers, and optimize regional transit routes.
- **Automated Invoicing Systems:** Development of a backend billing engine that automatically aggregates trip data, calculates tariffs, generates digital invoices, and tracks payment statuses.
- **Regional Analytics Engines:** Design of a data-processing pipeline capable of aggregating regional operational metrics to generate reports on fleet efficiency, delivery delays, and resource utilization.

2.4.2 Out of Scope

- **Hardware Sourcing and Manufacturing:** Procurement, manufacturing, or low-level firmware engineering of physical GPS tracking hardware modules or telematics equipment.

- **Telecommunications Infrastructure:** Maintenance, configuration, or operational management of cellular provider towers, cellular networks, or base stations.
- **Direct Financial Clearing Networks:** Development of localized transaction-clearing software, banking ledger infrastructure, or direct credit card processing networks (relying instead on external third-party payment gateways).

Part C: Problem Domain

The Problem Domain section indicates the operational realities of ABC-Trans and translates them into rigorous software engineering requirements by systematically analyzing the structural bottlenecks and data synchronization vulnerabilities inherent in the legacy systems. In addition, this part highlights the Task and Support framework to explicitly map core user-driven workflows against advanced architectural mechanisms, thereby suggesting robust solutions for real-time high concurrency, data integrity, and cross-branch synchronization across the newly unified enterprise ecosystem.

3.1 Pain Points

- **Operational Fragmentation:** ABC-Trans lacks extreme efficiency during daily working operations because of a heavy reliance on physical paperwork, slow manual data entry, and the absence of a unified system to connect online orders with in-branch workflows.
- **Lack of Branch Synchronization:** The reliance on isolated regional branch software and local databases leaves systems disconnected, trapping vehicle logs, driver assignments, and customer orders in individual silos without company-wide sharing.
- **Asset Scheduling Errors and Booking Conflicts:** Frequent fleet shortages at busy distribution hubs alongside system errors that cause truck double-bookings and overlapping driver schedules due to a lack of shared multi-user systems.
- **Visibility Gaps and Lack of Real-Time Tracking:** Total absence of live GPS tracking systems, forcing staff to rely on manual phone calls with drivers and preventing the system from providing accurate arrival time updates to partner supermarkets.
- **Slow Financial Processing and Billing Inefficiencies:** Delayed invoicing cycles and a high risk of human errors in billing files because delivery data is completely separated from back-office accounting platforms and manual payment cross-referencing.

3.2 Domain Entities

- **Branch:** Presents the regional distribution hubs and office locations across the nationwide network to manage localized resource allocation.
- **Order:** Represents the B2B corporate shipping requests submitted by partner supermarkets, containing cargo details, delivery destinations, and fulfillment status.

- **Vehicle:** Tracks the physical assets within the fleet, including truck metrics, real-time availability, and localized branch assignments.
- **Driver:** Manages the personnel records, active schedules, and compliance timelines including driver certification and license validity statuses.
- **Shipment:** Handles the active transit lifecycle of an order, linking specific vehicles and drivers to designated delivery routes.
- **Telemetry Log:** Captures real-time GPS coordinates and time telemetry data streams to provide live cargo tracking and accurate arrival updates.
- **Invoice:** Compiles the automated financial records, billing calculations, and transparent transaction receipts generated upon successful delivery fulfillment.
- **Payment:** Stores transaction histories, payment methods, and reconciliation statuses to ensure unalterable financial logging.

3.3 Actors

- **System Administrator:** Oversees the entire SmartFM platform, manages user accounts, configures branch settings, and maintains overall system security and data backups.
- **Branch Manager:** Monitors localized branch operations, reviews regional delivery reports, and handles local resource exceptions or disputes.
- **Dispatcher:** Assigns daily shipping orders to available vehicles, schedules driver shifts, and manages real-time route adjustments.
- **Driver:** Accesses the mobile interface to view assigned delivery schedules, updates shipment progress milestones, and reports delivery fulfillment statuses.
- **Accountant:** Manages the automated invoicing pipeline, cross-references payment reconciliation files, and handles billing discrepancies.
- **Partner Supermarket (Corporate Client):** Logs into the client portal to submit bulk B2B shipping requests, tracks active cargo locations, and views estimated arrival times.
- **External GPS Telemetry System:** An external hardware integration system that automatically streams real-time truck location coordinates and time tracking logs into the database.
- **External Accounting System:** A third-party financial platform that exchanges billing data and payment status updates with SmartFM to ensure automated ledger reconciliation.

3.4 Tasks

- **User and Access Management:**
 - ◆ **Actor:** System Administrator
 - ◆ **Description:** Manages employee accounts, configures role-based access controls, and updates system security permissions to protect company data.
- **Branch and Hub Setup:**
 - ◆ **Actor:** System Administrator, Branch Manager
 - ◆ **Description:** Configures regional distribution hub parameters, local database synchronization settings, and geographical boundaries for resource mapping.

- Fleet Asset Management:
 - ◆ Actor: System Administrator, Branch Manager
 - ◆ Description: Registers and updates physical transport assets, inputs vehicle technical specifications, and tracks localized truck availability.
- Order Management:
 - ◆ Actor: Partner Supermarket (Corporate Client), Dispatcher
 - ◆ Description: Handles the submission of bulk B2B shipping requests, details cargo dimensions, and logs specific delivery destination requests.
- Shipment Scheduling and Dispatch:
 - ◆ Actor: Dispatcher, Driver
 - ◆ Description: Assigns pending logistics orders to available trucks, schedules driver duty shifts, and resolves real-time routing changes.
- Real-Time Transit Tracking:
 - ◆ Actor: Driver, External GPS Telemetry System, Dispatcher
 - ◆ Description: Captures and records live GPS coordinates automatically while allowing drivers to log transit milestones manually.
- Handle Delivery Exceptions:
 - ◆ Actor: Driver, Dispatcher
 - ◆ Description: Manages unexpected transit issues such as vehicle breakdowns or traffic delays, allowing staff to reroute assets and notify supermarkets immediately.
- Process Order Modifications and Cancellations:
 - ◆ Actor: Partner Supermarket (Corporate Client), Dispatcher
 - ◆ Description: Processes real-time updates or cancellations of submitted orders before dispatch, updating vehicle allocation statuses instantly.
- Automated Invoice Generation:
 - ◆ Actor: Accountant, External Accounting System
 - ◆ Description: Compiles automated financial summaries immediately upon delivery completion, calculating precise shipping costs and fees.
- Payment Reconciliation:
 - ◆ Actor: Accountant, External Accounting System
 - ◆ Description: Cross-references automated billing files with third-party transaction logs to confirm payments and resolve invoicing errors.

Part D: Functional Requirements and Task Descriptions

4.1 User and Access Management

Task: User and Access Management	
Attribute	Description
Purposes	➤ Manage employee accounts: create, read, update, or deactivate user profiles. ➤ Enfore security boundaries: establish role-based access control and manage permission settings
Trigger/ Precondition	➤ A new employee joins ABC-Trans, an existing staff member changes corporate roles, or a security policy updates.
Frequency	➤ Low to variable, depending on corporate hiring cycles, internal role reassignments, and administrative updates.
Critical	➤ Securely manage system access and safeguard confidential corporate logistics data from unauthorized personnel
Work Area	➤ Administrative Dashboard
Sub-Tasks	Example Solution (Funtional Requirement)
Manage User Account	➤ Provide clean interfaces to input employee details: full name, corporate email, localized branch assignment, and active status. ➤ Ensure data integrity during profile changes.
Configure Access Permissions	➤ Automatically assign predefined feature permissions based on the designated corporate role. ➤ Allow manual permission overrides by authorized administrators as needed.
Variant	Example Solution (Funtional Requirement)
Integration with Central Directory	➤ The user management database syncs with internal company directory listings for unified employee credential validation
Automated Session Logging	➤ The system logs administrative account modifications automatically to maintain an unalterable security audit trail.

4.2 Branch and Hub Setup

Task: Branch and Hub Setup	
Attribute	Description
Purposes	➤ Establishes and configures regional distribution hub profiles within the centralized network.. ➤ Set localized database synchronization parameters to ensure cross-branch data consistency. ➤ Define geographic service boundaries and operating constraints for individual branches
Trigger/ Precondition	➤ A new regional distribution center is opened, an existing hub updates its infrastructure, or network synchronization protocols are modified by management
Frequency	➤ Low, occurring primarily during initial system deployment, geographic expansion, or corporate infrastructure reconfigurations.
Critical	➤ Highly critical to eliminate information silos, guarantee correct regional asset allocation, and maintain real-time data integrity across the entire corporate network.
Work Area	➤ Administrative Control Panel / System Configuration Dashboard
Sub-Tasks	Example Solution (Funtional Requirement)
Register Regional Hub	➤ Provide clean interfaces to input branch details including unique hub ID, geographic location, manager profile links, and storage capacities. ➤ Implement dynamic validation checks to prevent overlapping branch identification codes or duplicated directory records
Map Georaphic Boundaries	➤ Provide interactive digital map tools to plot specific regional service boundaries and delivery coverage areas for each hub. ➤ Automatically save spatial coordinate boundaries into the central database to guide automated order allocation systems.
Configure Synchronization Protocols	➤ Allow administrators to define automated database replication schedules, synchronization intervals, and data transmission bandwidth limits. ➤ Provide manual triggers that allow authorized branch managers to force immediate data synchronization updates during emergencies.

Monitor Network Connectivity	➤ Provide a centralized monitoring dashboard that displays the active connection status and latency metrics of all regional branch databases. ➤ Generate automated system alerts to notify technical personnel immediately if a regional node experiences connectivity drops or replication failures.
Variant	Example Solution (Functional Requirement)
Offline Operational Mode	➤ The regional branch database switches to an isolated offline mode during network outages, caching local transactions securely and queueing them for automatic upload once connectivity is restored.
Cross-Hub Resource Sharing	➤ The system provides temporary data viewing permissions that allow neighboring hubs to review shared asset availability logs during localized vehicle shortages.

4.3 Fleet Asset Management

Task: Fleet Asset Management	
Attribute	Description
Purposes	➤ Register and categorize physical transport vehicles within the centralized logistics network. ➤ Monitor real-time vehicle availability statuses to prevent asset scheduling conflicts. ➤ Schedule and track preventative maintenance routines to ensure fleet safety and operational compliance.
Trigger/ Precondition	➤ The company acquires new transport assets, an active vehicle requires maintenance or a truck is reassigned to a different regional distribution hub.
Frequency	➤ High, with daily real-time status updates alongside periodic maintenance inputs and vehicle registrations.
Critical	➤ Highly critical to optimize asset utilization, prevent scheduling conflicts, guarantee transport safety, and match cargo dimensions perfectly with vehicle capacities.
Work Area	➤ Fleet Management Dashboard (Operations Portal)
Sub-Tasks	Example Solution (Functional Requirement)

Register Transport Assets	➤ Provide standardized data input forms to record vehicle identification numbers, license plates, fuel types, and ownership documentation. ➤ Enforce strict parameter validation rules for maximum payload weight boundaries and total volume capacity configurations.
Track Asset Operational Status	➤ Automatically update vehicle availability profiles across distinct operational states including Available, En Route, In Maintenance, and Out of Service. ➤ Provide visual status alerts to notify dispatchers instantly when a vehicle becomes unavailable for daily assignment.
Schedule Fleet Maintenance	➤ Provide tracking logs to record mechanical inspections, repair histories, parts replacements, and insurance renewal dates. ➤ Generate automated notification alerts when a vehicle approaches its scheduled odometer mileage limit or time-based maintenance deadline.
Allocate Assets to Branches	➤ Allow authorized system administrators to assign or transfer specific vehicles to regional distribution hubs based on localized market demands. ➤ Maintain a centralized ledger tracking historical branch assignments and fleet migration logs across the entire corporation.
Variant	Example Solution (Funtional Requirement)
Cold-Chain Telemetry Integration	➤ The system integrates with specialized refrigerated container sensors to stream live internal temperatures, alerting drivers and dispatchers instantly if temperature levels deviate from supermarket safety standards.
Odometer and Fuel Log Automation	➤ The system automatically syncs with vehicle computer logs to upload mileage data and calculate average fuel consumption rates after each completed shipping delivery.

4.4 Order Management

Task: Order Management	
Attribute	Description
Purposes	➤ Provide corporate clients with centralized tools to submit and track bulk shipping requests. ➤ Validate cargo dimensions and physical constraints automatically against available fleet profiles. ➤ Maintain an organized digital repository of all pending, active, and completed B2B logistics orders.
Trigger/ Precondition	➤ A partner supermarket logs into the corporate client portal to upload a new shipment request, or an authorized dispatcher manually inputs order adjustments.
Frequency	➤ High, operating continuously throughout daily business hours as supermarkets restock inventory.
Critical	➤ Highly critical to capture accurate delivery data, eliminate manual order entry errors, and ensure appropriate cargo capacity planning before scheduling.
Work Area	➤ Corporate Client Portal (Order Processing Dashboard)
Sub-Tasks	Example Solution (Functional Requirement)
Submit Bulk Cargo Requests	➤ Provide a file upload interface within the client portal that allows partner supermarkets to import bulk orders using standardized digital spreadsheets. ➤ Provide structured data fields for manual single-order entry, capturing shipper details, pickup locations, drop-off destinations, and target delivery windows.
Validate Order Specifications	➤ Automatically check input values to ensure total cargo weight, volume dimensions, and special storage requirements conform to standard system limits. ➤ Cross-reference destination addresses instantly with regional branch databases to assign ownership to the correct distribution hub.
Track Order Lifecycles	➤ Dynamically transition order visibility profiles across standardized operational states including Pending, Approved, Dispatched, In Transit, and Delivered. ➤ Allow both dispatchers and partner supermarkets to filter, search, and view historical order records using unique tracking identification codes.

Calculate Preliminary Estimates	➤ Automatically generate initial shipping cost calculations and estimated arrival timelines based on distance matrices and cargo volume configurations. ➤ Display these baseline summaries to corporate clients immediately upon successful order submission for internal corporate tracking..
Variant	Example Solution (Funtional Requirement)
Automated Recurring Orders	➤ The system allows supermarkets to schedule automated recurring delivery templates for standard weekly inventory replenishment routes, reducing repetitive manual submissions.
API Order Ingestion Channel	➤ The system exposes secure external application programming interfaces that allow enterprise supermarket database infrastructures to stream order data directly into SmartFM without manual file uploads.

4.5 Shipment Scheduling and Dispatch

Task: Shipment Scheduling and Dispatch	
Attribute	Description
Purposes	➤ Assign pending logistics orders to verified available vehicles and qualified drivers. ➤ Optimize transit routes dynamically to reduce delivery delays and fuel consumption. ➤ Prevent booking duplication errors by locking assets immediately upon dispatch confirmation.
Trigger/ Precondition	➤ New validated orders are ready in the system pool, a dispatcher initiates a scheduling cycle, or an unexpected change requires route reconfiguration.
Frequency	➤ High, operating continuously as dispatchers clear the order queue and manage daily driver shifts.
Critical	➤ Highly critical to maximize fleet utilization, meet strict corporate supermarket delivery deadlines, and eliminate double-booking errors.
Work Area	➤ Central Dispatch Control Panel (Interactive Scheduling Board).
Sub-Tasks	Example Solution (Funtional Requirement)

Match Orders with Assets	➤ Provide a centralized interface listing unassigned orders alongside active, available trucks and drivers within the local branch database. ➤ Enforce automated verification rules to ensure the selected vehicle possesses the required capacity metrics for the assigned cargo load.
Lock Dispatched Resources	➤ Automatically transition the operational status of selected trucks and drivers to En Route immediately upon dispatch confirmation. ➤ Restrict locked assets from appearing in other active assignment pools across all branches to eliminate double-booking errors.
Generate Route Planning	➤ Provide interactive map integration that automatically calculates the most efficient delivery sequence based on distance matrices and destination points. ➤ Allow dispatchers to manually adjust stops or reorder delivery points on the digital map to accommodate specific regional constraints.
Transmit Trip Details	➤ Automatically push detailed trip sheets including pickup logs, customer details, delivery windows, and optimized routes directly to the designated driver mobile interface. ➤ Log dispatch execution timestamps within the central database to establish a clear audit trail for operational performance.
Variant	Example Solution (Funtional Requirement)
Automated Fleet Recommendation	➤ The system utilizes internal heuristics to automatically suggest the most cost-effective vehicle and driver combination for a group of orders based on proximity and capacity.
Multi-Leg Dispatch Ingestion	➤ The system supports complex multi-stop routing configurations, allowing a single vehicle to fulfill shipments for multiple partner supermarkets within a single dispatch cycle.

4.6 Real-Time Transit Tracking

Task: Real-Time Transit Tracking	
Attribute	Description
Purposes	➤ Capture continuous live location coordinates from active delivery vehicles automatically ➤ Provide supervisors and partner supermarkets with interactive geographic visibility of cargo progress. ➤ Establish an unalterable history of route deviations and milestone completion timestamps.
Trigger/ Precondition	➤ A driver initiates an assigned delivery trip on the mobile application, activating the vehicle tracking pipeline and hardware telemetry stream.
Frequency	➤ Continuous, streaming location updates every few seconds while a transport asset remains actively in transit.
Critical	➤ Highly critical to eliminate operational visibility gaps, guarantee cargo security, and provide partner supermarkets with precise estimated arrival times.
Work Area	➤ Live Tracking Map (Monitoring Control Center / Driver Mobile Interface)
Sub-Tasks	Example Solution (Functional Requirement)
Ingest Telemetry Streams	➤ Automatically receive continuous GPS latitude and longitude coordinate inputs from the External GPS Telemetry System hardware. ➤ Parse incoming data streams instantly to extract vehicle identification keys, current velocity metrics, and exact time logs.
Render Interactive Tracking Map	➤ Display active vehicle positions dynamically onto a centralized digital map using distinct color-coded truck icons based on transit statuses. ➤ Project the planned route geometry alongside the actual traveled path to help dispatchers identify route deviations instantly.
Log Delivery Milestones	➤ Provide simple touchscreen interfaces for drivers to log major transit checkpoints including Arrived at Hub, Loading, In Transit, and Completed. ➤ Automatically capture the exact system timestamp and spatial coordinates whenever a driver toggles a milestone button.

Calculate Estimated Arrival Times	➤ Compute real-time estimated arrival duration updates periodically by checking current transit progress against traffic congestion metrics and distance vectors. ➤ Expose these estimated arrival timelines directly onto the partner supermarket portal to keep corporate clients informed.
Variant	Example Solution (Functional Requirement)
Geofencing Verification Alerts	➤ The system establishes automated digital perimeters around target supermarket warehouses, triggering instant arrival alerts to facility managers the moment a delivery truck crosses the geographic boundary.
Telemetry Loss Data Caching	➤ The driver mobile application tracks location points independently during cellular coverage dropouts, securely caching coordinates locally and pushing them to the central server immediately upon network restoration.

4.7 Handle Delivery Exceptions

Task: Handle Delivery Exceptions	
Attribute	Description
Purposes	➤ Provide immediate communication channels for reporting unexpected transit incidents. ➤ Enable operational staff to execute rapid contingency plans and reroute logistics assets.. ➤ Maintain documented evidence of transit disruptions for corporate liability verification.
Trigger/ Precondition	➤ A driver encounters a mechanical failure, traffic blockage, or cargo discrepancy, or a supermarket rejects a delivery due to product issues.
Frequency	➤ Low to medium, occurring dynamically whenever operational anomalies disturb normal transit schedules.
Critical	➤ Highly critical to minimize delivery delays, preserve corporate service level agreements, and ensure rapid vehicle replacement during emergencies.
Work Area	➤ Driver Mobile Application (Dispatcher Incident Management Board).
Sub-Tasks	Example Solution (Functional Requirement)

Report Active Incidents	➤ Provide a simplified interface on the mobile application for drivers to select incident categories such as Vehicle Breakdown, Accident, Severe Traffic, or Cargo Damage. ➤ Allow drivers to attach smartphone photos and text descriptions directly to the incident report before submission.
Broadcast Emergency Alerts	➤ Automatically transmit high-priority warning notifications to the central dispatcher dashboard the moment a driver submits an exception event. ➤ Flag the affected transport vehicle on the interactive live map with a distinctive visual warning indicator to denote emergency status.
Execute Asset Rerouting	➤ Provide dispatchers with real-time tools to detach remaining orders from a disabled vehicle and assign them to the nearest available truck. ➤ Instantly push the newly modified route data and updated trip sheets to the backup driver mobile application.
Notify Affected Supermarkets	➤ Automatically generate and send system notifications to the target partner supermarket portal regarding the delayed delivery status. ➤ Provide updated estimated arrival calculations that account for the time lost during the incident resolution process.
Variant	Example Solution (Functional Requirement)
Partial Cargo Rejection Routine	➤ If a partner supermarket rejects specific damaged cargo items upon arrival, the system allows the driver to modify delivery quantities on-site, capturing digital signatures and initiating an immediate partial return shipment task.
Automated Towing Service Dispatch	➤ For critical mechanical failures, the system cross-references vehicle coordinates with pre-registered corporate service partners to automatically recommend nearby maintenance hubs or towing companies to the dispatcher.

4.8 Process Order Modifications

Task: Process Order Modifications	
Attribute	Description
Purposes	➤ Allow partner supermarkets to submit real-time update or cancellation requests for active orders. ➤ Recalculate vehicle capacity allocations dynamically when modifications are confirmed. ➤ Prevent wasted fleet resources by instantly freeing up vehicles assigned to canceled shipments.
Trigger/ Precondition	➤ A partner supermarket requests an adjustment via the client portal, or a dispatcher manually modifies order entries prior to physical vehicle dispatch.
Frequency	➤ Medium, occurring dynamically based on supermarket inventory fluctuations or scheduling shifts.
Critical	➤ Highly critical to maintain flexible logistics operations, maximize vehicle capacity utilization, and avoid empty or redundant transport journeys.
Work Area	➤ Corporate Client Portal (Dispatcher Order Dashboard).
Sub-Tasks	Example Solution (Functional Requirement)
Receive Modification Requests	➤ Provide clear interface forms within the client portal for supermarkets to alter shipment details including cargo quantities, drop-off locations, or delivery times. ➤ Implement strict system blockades that automatically reject modification requests once the designated vehicle has passed the physical dispatch checkpoint.
Validate Altered Capacities	➤ Automatically re-evaluate cargo weight and volume adjustments against the maximum payload configurations of the assigned transport truck. ➤ Trigger instant warning messages to dispatchers if the requested order expansion exceeds the current available space of the scheduled vehicle.
Process Order Cancellations	➤ Allow authorized corporate clients to cancel pending shipments directly through the portal before route execution begins. ➤ Automatically wipe the canceled order ties from the dispatch schedule, reverting the vehicle and driver statuses back to Available instantly.

Log Modification Histories	➤ Maintain a complete digital record of all order changes, capturing original values, updated details, user identification keys, and processing timestamps. ➤ Highlight modified orders on the central dispatcher control board with distinct visual markers to ensure staff awareness.
Variant	Example Solution (Functional Requirement)
Automated Fee Calculation	➤ The system automatically applies predefined corporate penalty fees or processing adjustments to the final client billing profile depending on how close the cancellation occurred to the scheduled dispatch time
Smart Route Recalculation	➤ For complex multi-stop deliveries, if an intermediate supermarket cancels its order, the system automatically bypasses that specific geographic stop and optimizes the remaining delivery sequence instantly.

4.9 Automated Invoice Generation

Task: Automated Invoice Generation	
Attribute	Description
Purposes	➤ Calculate precise B2B shipping fees automatically upon delivery completion. ➤ Generate digital electronic invoices based on contractual corporate pricing structures. ➤ Reduce financial processing cycle times by eliminating manual data translation errors.
Trigger/ Precondition	➤ A driver successfully updates an order status to Delivered on the mobile application, or a dispatcher manually confirms a successful trip completion.
Frequency	➤ High, executing automatically in real time whenever an active logistics delivery cycle concludes.
Critical	➤ Highly critical to guarantee transparent corporate billing, secure revenue streams, and maintain financial data integrity for auditing purposes.
Work Area	➤ Financial Accounting Portal (Automated Billing System).
Sub-Tasks	Example Solution (Functional Requirement)

Extract Trip Performance Data	➤ Automatically retrieve completed delivery metrics including total traveled distance logs, validated cargo weight values, and specific stop counts from the tracking database. ➤ Link the extracted operational metrics directly to the corresponding corporate partner account profile.
Execute Cost Calculation	➤ Compute baseline shipping fees dynamically by applying pre-negotiated corporate tariff matrix parameters stored within the client contract configuration. ➤ Automatically incorporate authorized extra charges including toll road expenses, waiting time penalties, or special cold-chain handling premiums.
Compile Electronic Invoices	➤ Generate standardized digital invoice files featuring unique billing reference numbers, itemized charge descriptions, tax calculations, and corporate payment deadlines. ➤ Store the newly compiled electronic documents securely within the central financial ledger database in an unalterable format.
Publish Financial Summaries	➤ Automatically push the generated electronic invoice files directly to the designated partner supermarket client portal for immediate review. ➤ Send automated email notifications containing summary billing attachments to pre-configured corporate accounting contact addresses.
Variant	Example Solution (Functional Requirement)
Consolidated Periodic Billing	➤ Instead of generating standalone invoices per delivery, the system aggregates all completed trip data for a specific supermarket chain over a weekly or monthly cycle to produce a single consolidated billing statement.
Dynamic Currency Conversion	➤ The system automatically checks active international exchange rate registries to convert and display invoice totals in multiple supported currencies for multinational enterprise clients.

4.10 Payment Reconciliation

Task: Payment Reconciliation	
Attribute	Description
Purposes	➤ Establish automated cross-referencing between internal billing invoices and external financial transaction logs. ➤ Detect and flag balance discrepancies or settlement errors immediately to protect corporate revenue. ➤ Update order accounting payment records automatically to reflect accurate operational cash flows.
Trigger/ Precondition	➤ External bank statements or transaction files are uploaded into the platform, a scheduled reconciliation batch cycle triggers, or an accountant initiates a manual audit review.
Frequency	➤ Periodic to continuous, typically executing automatically at the end of each business day or weekly operational cycle.
Critical	➤ Highly critical to eliminate manual financial auditing mistakes, prevent cash leakage, and provide clean financial statement tracking for enterprise auditing.
Work Area	➤ Accountant Audit Dashboard / Financial Management Center
Sub-Tasks	Example Solution (Functional Requirement)
Ingest Transaction Logs	➤ Provide secure data channels to import electronic payment statements and digital banking files containing transaction dates, payment references, and values. ➤ Parse incoming financial records automatically to extract data fields while keeping secure encryption controls active.
Execute Matching Rules	➤ Run internal matching heuristics to compare statement transactions against outstanding electronic invoices using unique order codes and billing total values. Automatically change the invoice payment profile state to Settled the moment a perfect financial data match is confirmed by the system.
Flag Discrepancies	➤ Automatically filter and redirect unmatched records, partial payments, or overpaid invoices into a dedicated system exception queue for administrative review. ➤ Highlight flagged discrepancies on the accountant dashboard with high-priority visual markers alongside calculated deficit amounts.

Generate Audit Summaries	➤ Compile comprehensive daily reconciliation reports detailing total invoice totals, confirmed deposits, remaining outstanding balances, and active dispute metrics. ➤ Export financial reconciliation summaries into standardized unalterable data files to assist with corporate tax or internal accounting compliance.
Variant	Example Solution (Functional Requirement)
Direct Banking API Link	➤ The system utilizes secure real-time bank application programming interfaces to fetch transaction feeds directly as they happen, executing immediate instant matching actions instead of relying on manual file uploads.
Automated Dispute Resolution	➤ For minor, acceptable payment variations within predefined financial tolerance thresholds, the system automatically posts balancing adjustments to clear the ledger while logging the deviation for history checks.

Part E: Workflow

5.1 User and Access Management

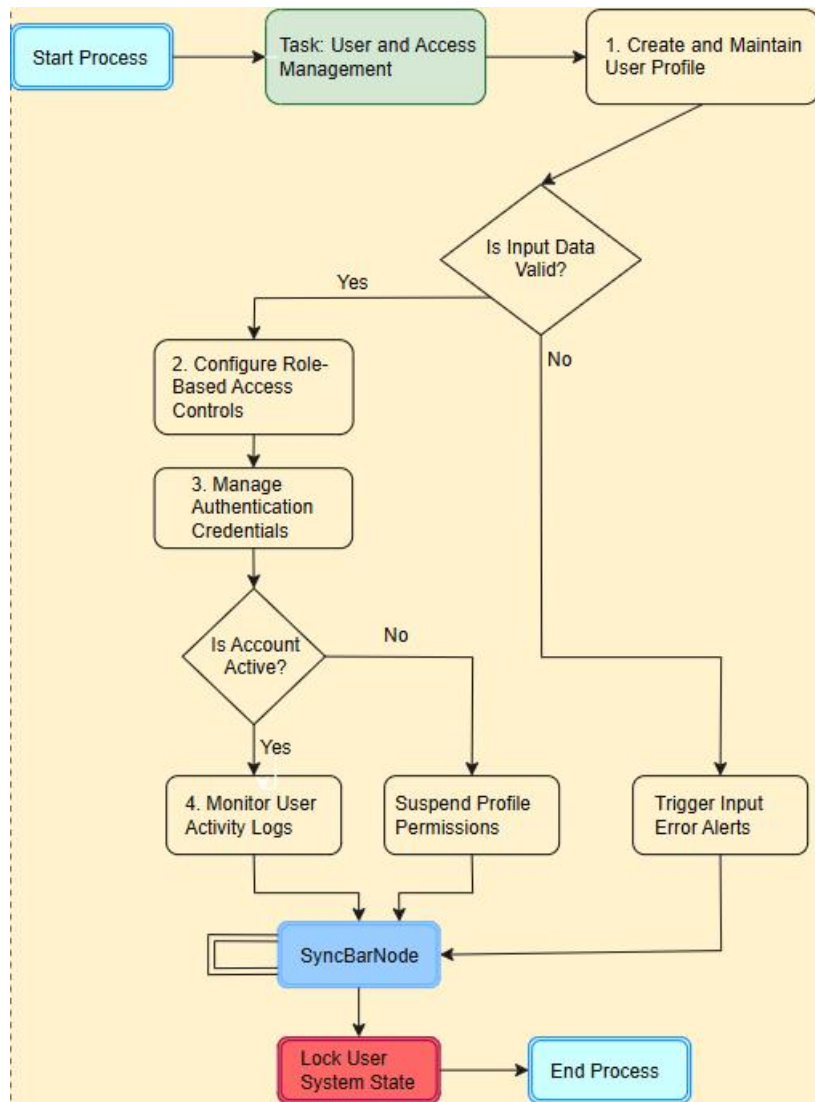


Figure 1: Workflow of User and Access Management Task

5.2 Branch and Hub Setup

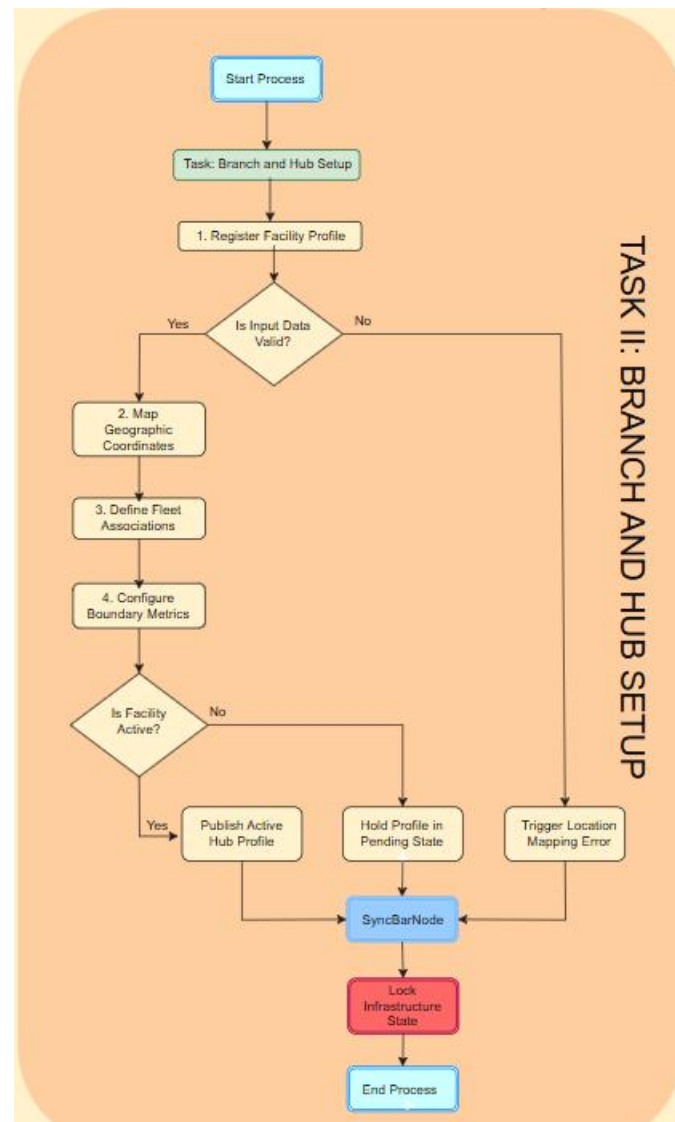


Figure 2: Workflow of Branch and Hub Setup Task

5.3 Fleet Asset Management

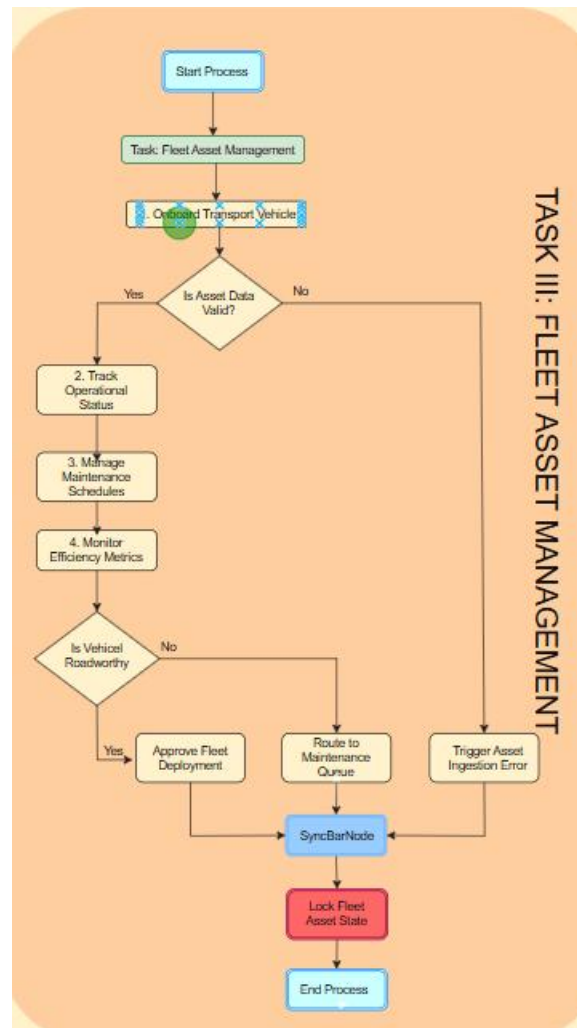


Figure 3: Fleet Asset Management Workflow

5.4 Order Management

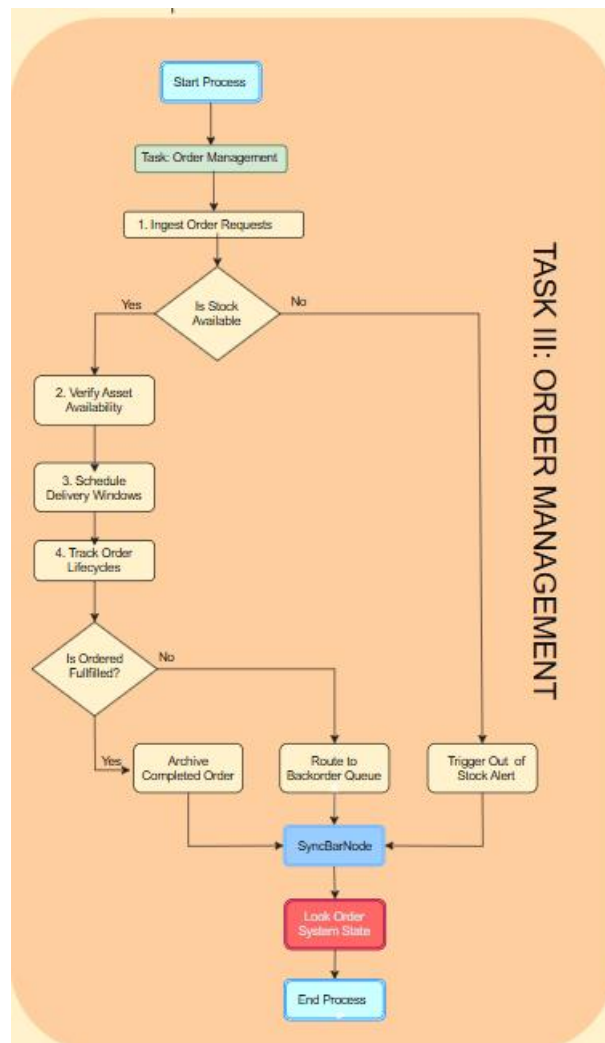


Figure 4: Order Management Workflow

5.5 Shipment Scheduling and Dispatch

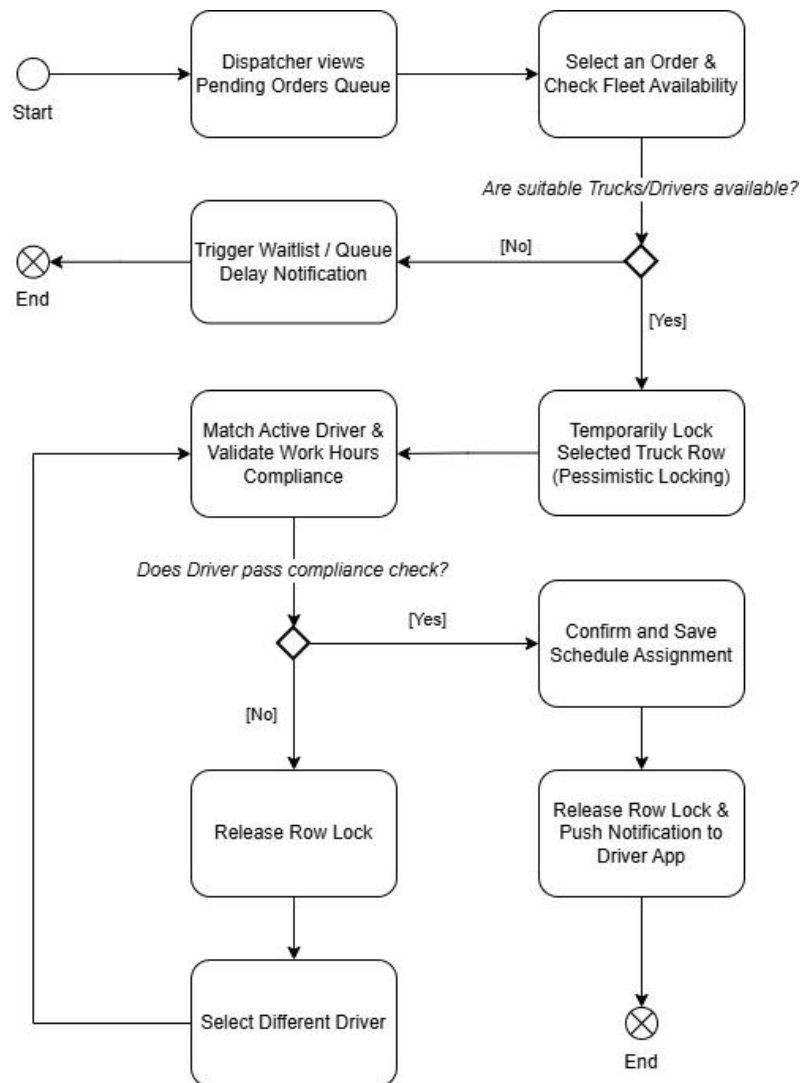


Figure 5: Shipment Scheduling and Dispatch Workflow

5.6 Real-Time Transit Tracking

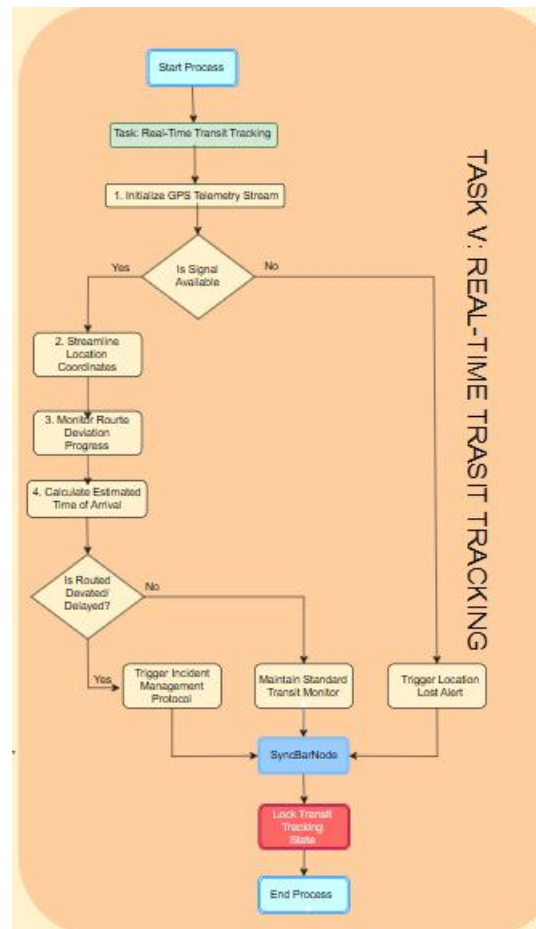


Figure 6: Real-Time Transit Tracking Workflow

5.7 Handle Delivery Exceptions

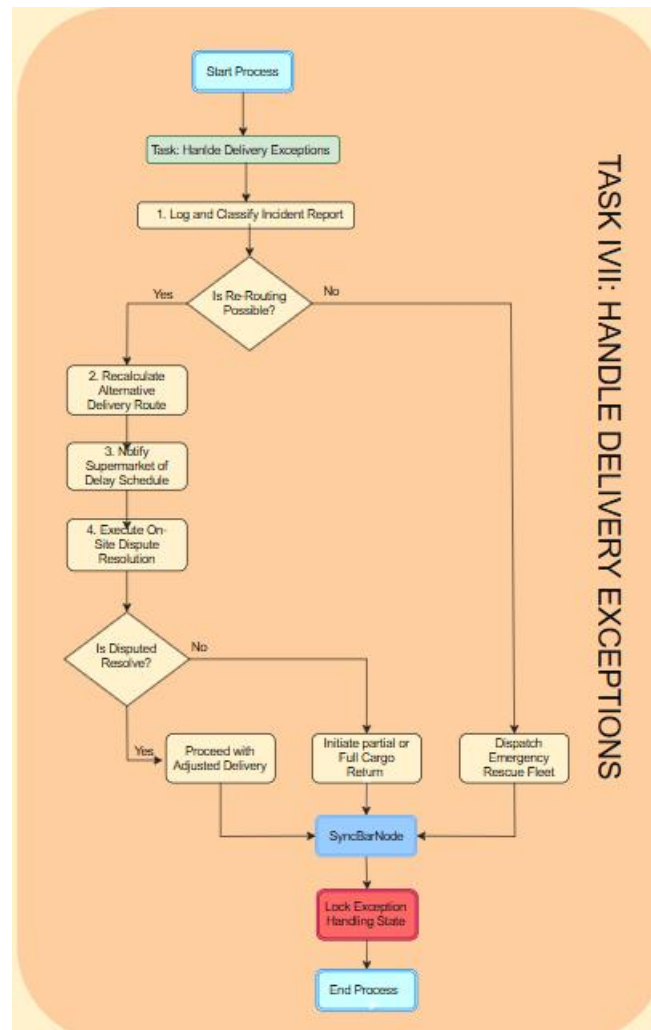


Figure 7: Handle Delivery Exceptions Workflow

5.8 Process Order Modifications

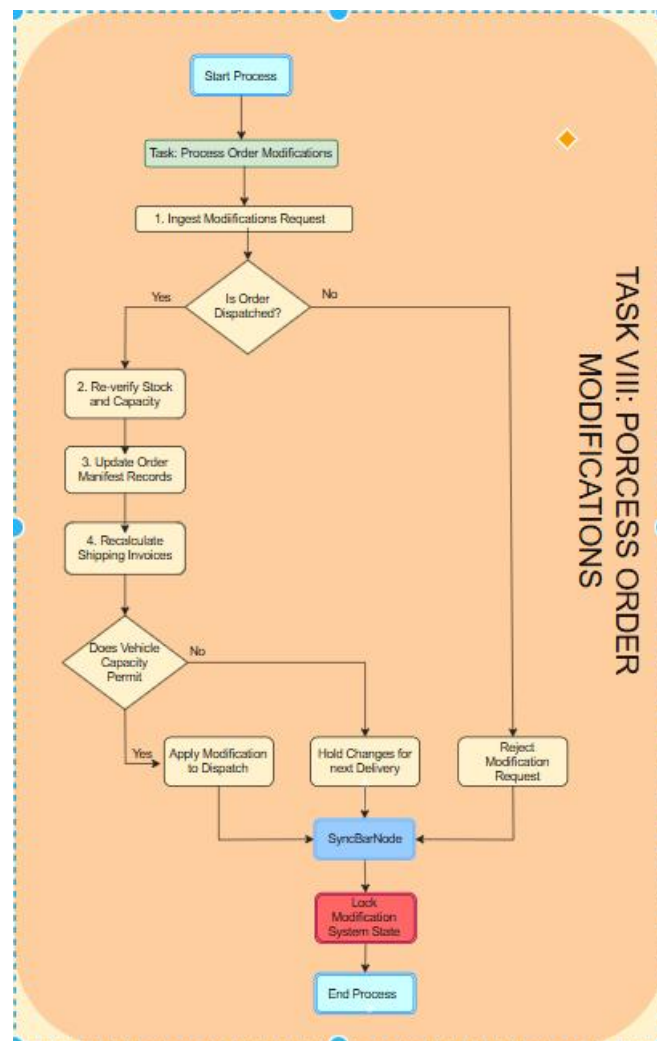


Figure 8: Process Order Modifications Workflow

5.9 Automated Invoice Generation

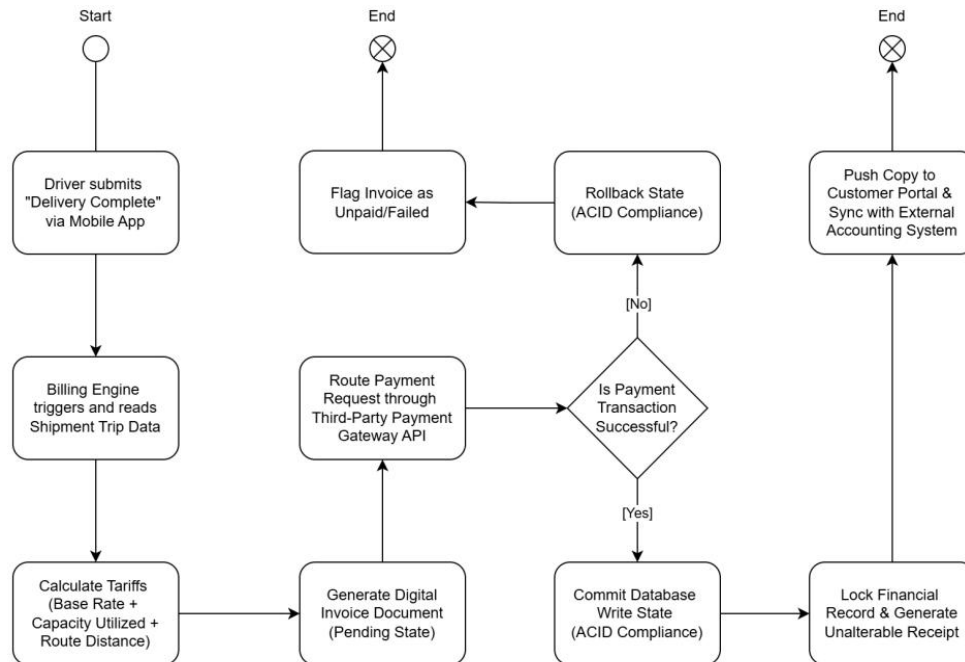


Figure 9: Automated Invoice Genration Workflow

5.10 Payment Reconciliation

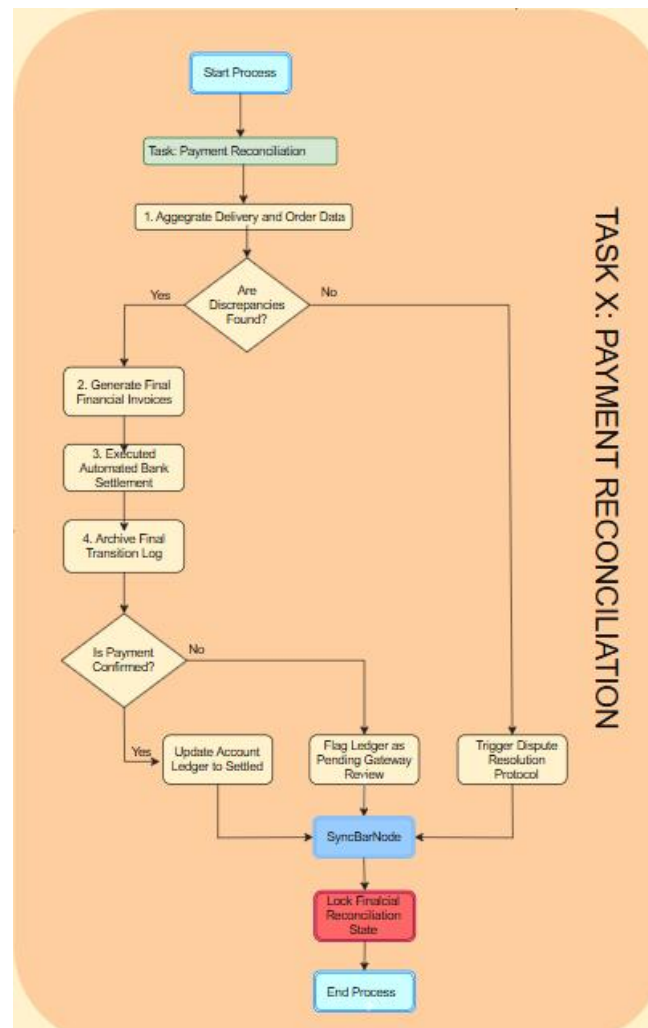


Figure 10: Payment Reconciliation Workflow

Part F: Data Model

After the detailed specification of operational workflow, this part highlights the structural foundation of the Smart Fleet Management platform through comprehensive Data Models. While previous workflows map the dynamic behavioral logic and chronological step-by-step processing of logistics events, the Data Model formalizes how critical operational entities are structured, stored, and integrated within the centralized database architecture. Therefore, a rigorous data design is paramount to guaranteeing data integrity, eliminating informational redundancy, and ensuring high-throughput transaction processing for ABC-Trans' complex supply chain operations.

6.1 Conceptual Data Model

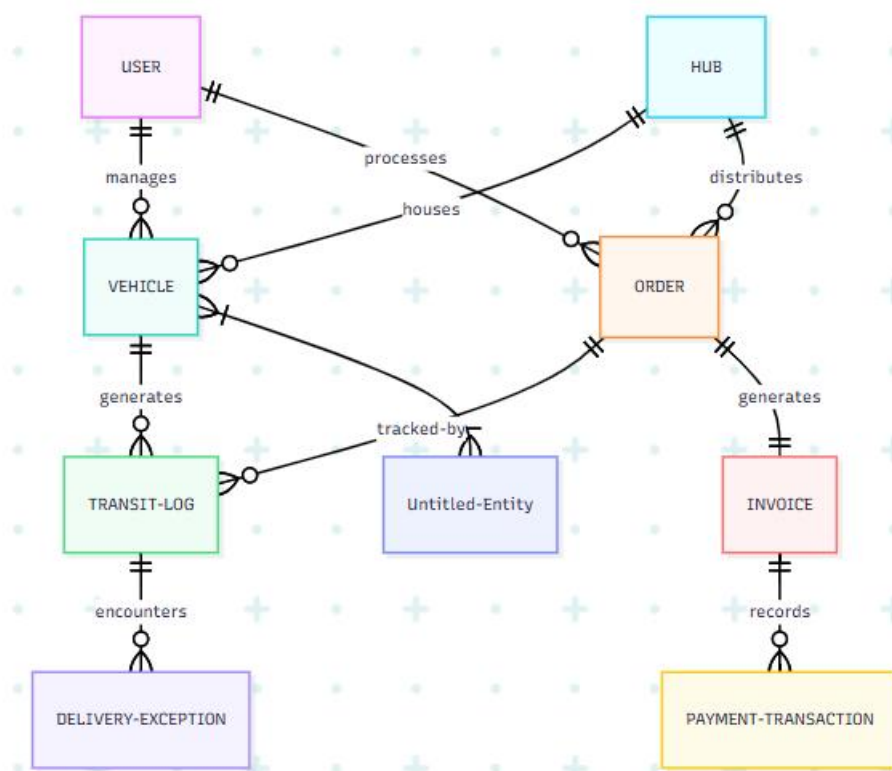


Figure 11: Conceptual Data Model

The following structural principles outline how the core entities within Figure 11 interact and maintain data relationship integrity:

- USER to VEHICLE Relationship (manages):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: A fleet manager monitors multiple transport assets during each truck must be assigned to exactly one specific manager to ensure clear accountability.
 - Data Constraints: The "||" symbol on the **USER** side represents a Mandatory One constraint, meaning a vehicle cannot exist without an assigned manager. Besides that, the "o{" symbol on the **VEHICLE** side represents an Optional Many constraint, allowing a new manager

to have zero vehicles initially but gradually take control of a large fleet over time.

- USER to ORDER Relationship (processes):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: An authorized dispatcher processes incoming supermarket requests to prepare them for physical logistical fulfillment.
 - Data Constraints: The "||" notation enforces that every commercial order is handled under the supervision of exactly one staff member. Furthermore, the "o{" notation indicates a dynamic workflow, where a user can have zero active orders during off-peak hours and handle hundreds during peak operations.
- HUB to VEHICLE Relationship (houses):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: Each regional hub houses a dedicated transport fleet, while every physical truck must be registered to exactly one primary home base.
 - Data Constraints: The "||" on the HUB side represents a Mandatory One constraint to prevent scheduling conflicts. Furthermore, the "o{" symbol on the VEHICLE side represents an Optional Many constraint, allowing a new hub to manage zero trucks initially and scale up over time.
- HUB to ORDER Relationship (distributes):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: A regional hub fulfills and dispatches delivery orders to supermarkets within its designated territory.
 - Data Constraints: The "||" notation requires each order to originate from exactly one hub to avoid duplicate shipping errors. Furthermore, the "o{" notation allows a hub to start the business day with zero pending requests but distribute a large volume of orders during peak hours.
- VEHICLE to TRANSIT-LOG Relationship (generates):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: A delivery vehicle streams its continuous telemetry and real-time location data directly into the tracking database.
 - Data Constraints: The "||" constraint ensures every GPS data stream maps back to exactly one accountable vehicle. Additionally, the "o{" constraint accounts for operational states, yielding zero logs when the truck is parked and thousands of logs during active routes.
- TRANSIT-LOG to DELIVERY-EXCEPTION Relationship (encounters):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: A specific active transit route captures and logs unexpected on-road incidents for corporate risk assessment
 - Data Constraints: The "||" notation permanently locks an incident report to that exact trip instance. Additionally, the "o{" symbol represents realistic operations where a standard run encounters zero exceptions, but a disrupted trip can accumulate multiple incident records.
- ORDER to INVOICE Relationship (generates):
 - Relationship Type: One-to-One (1:1)

- Business Logic: Every completed store replenishment order initiates a compliance billing process to document the commercial transaction.
- Data Constraints: The link uses double vertical bars "||" on both ends, enforcing a strict mandatory One-to-One constraint to ensure an invoice cannot exist without a valid parent order.
- INVOICE to PAYMENT-TRANSACTION Relationship (records):
 - Relationship Type: One-to-Many (1:N)
 - Business Logic: A financial invoice tracks the electronic cash flow and banking history required to settle a supermarket bill.
 - Data Constraints: The "||" constraint links the transaction log to a single financial statement. Therefore, the o{ symbol accommodates digital banking realities, starting at zero records initially and scaling to multiple separate attempts if gateway timeouts occur.

6.2 Logical Data Model and Entity Relationship Diagram

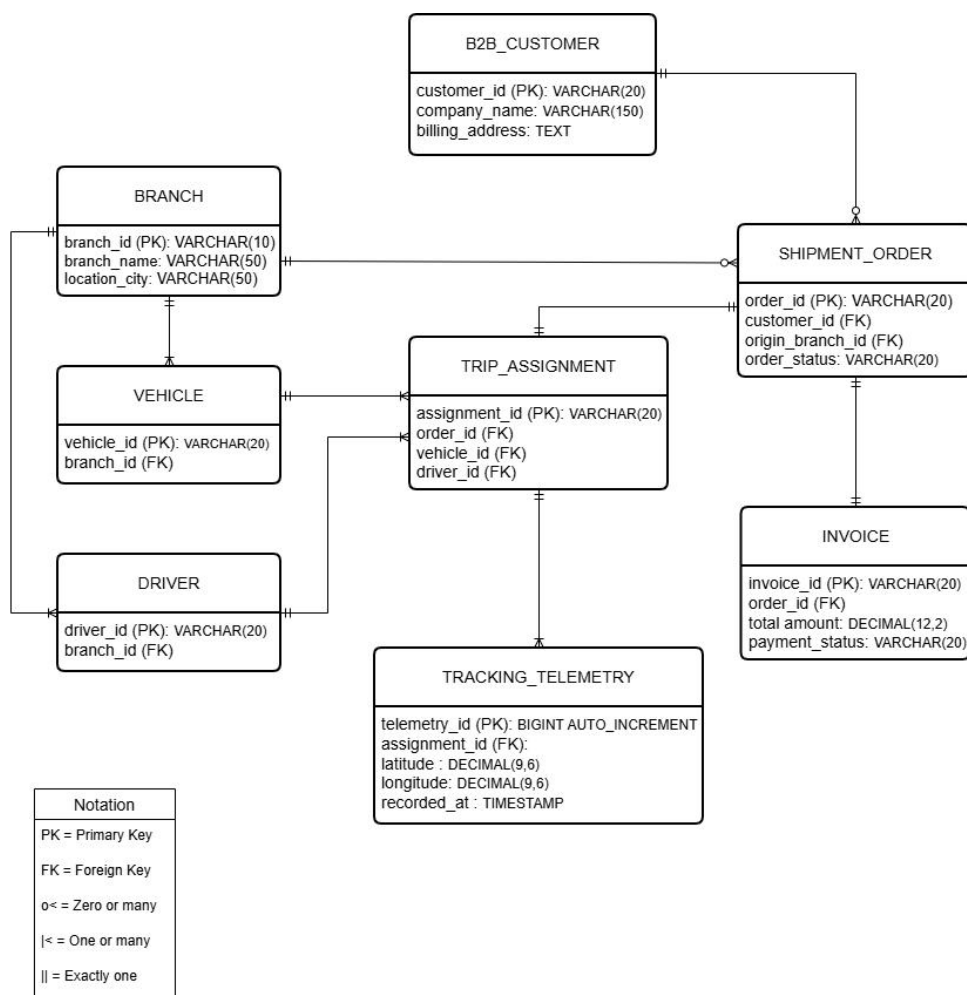


Figure 12: Detailed Entity Relationship Diagram (ERD) of SmartFM Platform

The **Figure 12** describes the attributes, primary keys (PK), and foreign keys (FK) for each core domain entity in the SmartFM system.

- VEHICLE
 - vehicle_id (PK) : VARCHAR(20) — Unique plate or fleet number.
 - branch_id (FK) : VARCHAR(10) — Current regional hub location.
 - truck_type: VARCHAR(30) — Rigid, Refrigerated, Container Truck.
 - weight_capacity: DECIMAL(10,2) — Maximum weight payload in tons.
 - volume_capacity : DECIMAL(10,2) — Cubic meter volume limitation.
 - maintenance_status: VARCHAR(20) — Available, In-Service, Maintenance.
 - lock_version: INT — Used for database row-level state tracking.
- DRIVER
 - driver_id (PK) : VARCHAR(20) — Unique employee license reference.
 - branch_id (FK) : VARCHAR(10) — Home branch assignment.
 - full_name : VARCHAR(100).
 - license_class: VARCHAR(10) — e.g., FC, E classes in Vietnam.
 - active_service_hours: DECIMAL(4,2) — Tracks compliance limits to prevent fatigue.
 - availability_status: VARCHAR(20) — Ready, On-Trip, On-Leave.
- SHIPMENT ORDER
 - order_id (PK) : VARCHAR(20)
 - customer_id (FK) : VARCHAR(20)
 - origin_branch_id (FK) : VARCHAR(10)
 - destination_address: TEXT — Target retail supermarket branch
 - cargo_weight : DECIMAL(10,2)
 - order_status: VARCHAR(20) — Pending, Dispatched, In-Transit, Fulfilled, Cancelled
 - created_at : TIMESTAMP
- TRIP ASSIGNMENT
 - assignment_id (PK) : VARCHAR(20)
 - order_id (FK) : VARCHAR(20) — Enforces strict 1-to-1 relationship mapping
 - vehicle_id (FK) : VARCHAR(20) — Protected by Pessimistic Locking during scheduling
 - driver_id (FK) : VARCHAR(20)
 - dispatch_time : TIMESTAMP
 - actual_delivery_time: TIMESTAMP (Nullable)
- TRACKING TELEMETRY (Decoupled write log for Read Replicas)
 - telemetry_id (PK): BIGINT AUTO_INCREMENT
 - assignment_id (FK) : VARCHAR(20)
 - latitude : DECIMAL(9,6) — Continuous GPS coordinate streams
 - longitude : DECIMAL(9,6)
 - recorded_at : TIMESTAMP
 - current_eta : TIMESTAMP
- INVOICE (ACID Compliance Protected)
 - invoice_id (PK) : VARCHAR(20)
 - order_id (FK) : VARCHAR(20)

- `base_tariff` : DECIMAL(12,2)
- `distance_surcharge` : DECIMAL(12,2)
- `total_amount` : DECIMAL(12,2)
- `payment_status`: VARCHAR(20) — Unpaid, Processing, Settled, Failed
- `issued_at` : TIMESTAMP

Part G: Quality Attributes of the System

As the ABC-Trans company shifts mainly on the system to drive its core logistics operations, the profound innovation generates a stringent set of non-functional requirements is imperative to guarantee operational viability and architectural resilience. As a result, the Quality Attributes of the System section highlights the core architectural benchmarks including the following sections required to sustain this digital transition:

7.1 Availability

The SmartFM platform must maintain a minimum uptime of 24/7 to prevent logistics disruptions for ABC-Trans, indicating that the architectural design is justified by three core pillars:

- **Business Continuity:** Real-time fleet coordination and shipment tracking cannot tolerate downtime. For instance, a 99.9% uptime ensures that order processing and hub distribution remain continuous without operational bottlenecks.
- **Multi-AZ Mitigation:** Deploying the application layer across multiple AWS Availability Zones removes single points of failure. Therefore, the alternate zones isolate the impact to maintain system stability if one data center encounters an outage.
- **Automated Failover:** Automated Elastic Load Balancers continuously monitor server health, instantly rerouting traffic away from compromised nodes to active instances with zero disruption to drivers or managers.

7.2 Performance and Scalability

The system processes high-throughput fleet telemetry and scales dynamically during operational surges. Specifically, the architectural design is justified by the three following requirements:

- **Latency Bounds:** Real-time GPS streams from active vehicle fleets must be processed with an end-to-end latency of less than 2 seconds. Consequently, this constraint guarantees that hub managers always view accurate vehicle positions during tight delivery windows.
- **Dynamic Auto Scaling:** To absorb seasonal commercial surges without service degradation, the backend microservices leverage AWS Auto Scaling groups which handling up to a 300% spike in concurrent traffic during peak holiday operations.

- **Database Offloading:** Amazon ElastiCache (Redis) caches frequent read operations, such as active branch profiles and driver configurations to keep the primary database free from read bottlenecks and maintain fast response times during heavy traffic.

7.3 Security and Privacy

In terms of Security and Privacy, the designed system must safeguard sensitive corporate data assets and maintain tight administrative control. Particularly, the architectural design is justified by the three following requirements:

- **Data Encryption:** All high-value systemic information including customer corporate billing details and user credentials must be protected using AES-256 encryption at rest and TLS 1.3 encryption in transit.
- **Role-Based Access:** The platform enforces strict Role-Based Access Control (RBAC) to ensure precise data isolation, preventing unauthorized roles such as field drivers from gaining visibility into sensitive financial invoices or system configurations.
- **System Governance:** Automated audit logging tracking all administrative actions must be securely stored, allowing security teams to review modification histories and prevent internal data leaks.

7.4 Maintainability

Next, in the Maintainability part, the architectural codebase must ensure long-term software viability and rapid feature evolution to support the operational growth of ABC-Trans including:

- **Clean Architecture:** The system separates core domain business logic from data access and external interface layers, leading decoupled approach prevents tight coupling, allowing developers to modify specific components without impacting the entire system state.
- **Low Technical Debt:** Adhering strictly to clean code paradigms ensures that the application logic reads like well-written prose and remains easy to understand. Furthermore, this structural clarity drastically reduces onboarding time for new developers and simplifies debugging.
- **Automated CI/CD Pipelines:** Continuous integration and deployment pipelines automatically trigger comprehensive unit testing suites before any production release. Thus, this automated validation guarantees system stability, allowing the team to deploy routine patches or hotfixes with zero downtime.

7.5 Reliability and Data Integrity

Then, regarding the Reliability and Data Integrity aspect, the system architecture must guarantee faultless execution under continuous operational stress and preserve the absolute accuracy of transactional records containing:

- **Database Transactions:** The platform strictly enforces ACID (Atomicity, Consistency, Isolation, Durability) compliance across all data storage layers that prevents partial data writes and ensures that the entire operational state is immediately rolled back if a database transaction fails midway.
- **Idempotent API Design:** To handle network disruptions or accidental double-submissions by users, all core backend APIs utilize strict idempotency tokens. In addition, our team believes that this design guarantees that duplicate requests such as multiple driver location updates and produces the exact same system outcome without altering existing data states.
- **Automated Backup Schedules:** Automated snapshots of the primary database are executed hourly, with copies safely replicated to remote storage buckets which ensures minimal data loss and enables rapid recovery in the event of a catastrophic infrastructure failure.

7.6 Fault Tolerance and Recoverability

Furthermore, focusing on the Fault Tolerance and Recoverability aspect, the underlying infrastructure must exhibit resilient self-healing capabilities and guarantee swift disaster recovery actions:

- **Circuit Breaker Patterns:** The microservices architecture integrates defensive programming design patterns to isolate internal component failures. If a non-critical third-party service slows down, the system trips the circuit, preventing localized latency from cascading into a total system blackout.
- **Dead Letter Queues:** Unprocessable application events or failed messaging payloads are automatically redirected to isolated storage queues, allowing the core operational pipeline to process subsequent traffic normally while administrators investigate the anomalous errors.
- **Disaster Recovery RTO and RPO:** The architectural deployment guarantees a Recovery Time Objective (RTO) of under 4 hours and a Recovery Point Objective (RPO) of less than 1 hour. Furthermore, these strict temporal thresholds ensure that operations can be safely restored with minimal business impact.

7.7 Usability

Finally, regarding the Usability aspect, the system interfaces must deliver intuitive, accessible, and high-efficiency operational interactions for all user segments across ABC-Trans, including:

- **Task-Oriented Dashboard Layouts:** The core interface design arranges informational components based on priority, allowing hub managers to complete critical dispatch tasks with minimal navigational steps. As a result, this visual hierarchy accelerates operational workflows and minimizes human entry errors.
- **Responsive Mobile Optimization:** The application interface scales seamlessly across various mobile screen dimensions, prioritizing high-

contrast UI elements and accessible touch targets. Therefore, this responsive optimization guarantees that field drivers can log tracking events comfortably under diverse outdoor working conditions.

- Onboarding Friction Reduction: The design utilizes self-explanatory interactive flows and contextual tooltips rather than relying on complex external instruction sheets. As my team expected, this native guidance shortens training times for new operational personnel and reduces overall system adoption friction.

Part H: Validation of Requirements

In the requirements validation section, a formal cross-reference verification is conducted to ensure that the specification for the Smart Fleet Management System (SmartFM) remains robust, cohesive, and entirely devoid of architectural gaps. Additionally, aligning with the industry-standard verification principles established by Swinsoft Consulting and applied to enterprise operations, the framework below establishes explicit traceability. As a result, this matrix systematically maps each core functional task from Part D against its structural data entities from Part F and its corresponding operational justification within the Non-Functional Quality Attributes of Part G, verified through a comprehensive Create, Read, Update, and Delete (CRUD) compliance architecture:

- Task 4.1: User and Access Management
 - Data Entities: USER, BRANCH
 - Quality Attribute: 7.3 Security and Privacy (*Role-Based Access Control*)
 - Validation Defense [C, R, U, D]: This binds user accounts strictly to designated enterprise roles. It stops local branch workers from messing with or viewing restricted financial metrics or global configurations across the whole system.
- Task 4.2: Branch and Hub Setup
 - Data Entities: BRANCH, HUB, VEHICLE
 - Quality Attribute: 7.1 Availability (*Multi-AZ Mitigation*)
 - Validation Defense [C, R, U]: This handles the creation and updating of regional distribution hubs. It keeps asset maps accurate and up to date across multiple cloud availability zones so the system stays up even if one zone fails.
- Task 4.3: Fleet Asset Management
 - Data Entities: VEHICLE, BRANCH, DRIVER
 - Quality Attribute: 7.2 Performance and Scalability (*Database Offloading*)
 - Validation Defense [C, R, U, D]: Truck profiles and their active statuses are cached directly inside Redis. This handles non-stop fleet availability lookups during dispatch without hammering the main database or bottlenecking writes when things get busy.
- Task 4.4: Order Management
 - Data Entities: ORDER, B2B_CUSTOMER, HUB
 - Quality Attribute: 7.7 Usability (*Task-Oriented Dashboards*)
 - Validation Defense [C, R, U, D]: Streamlines bulk ingestion setups for our partner supermarkets. The clean, focused dashboard UI reduces human typing mistakes when orders flood the system during peak shipping hours.

- Task 4.5: Shipment Scheduling and Dispatch
 - Data Entities: TRIP_ASSIGNMENT, VEHICLE, DRIVER, ORDER
 - Quality Attribute: 7.5 Reliability and Data Integrity (*Database Locks*)
 - Validation Defense [C, R, U]: Uses a strict pessimistic row-level locking strategy via a lock_version check. This prevents two different dispatchers from assigning the same truck or driver to different trips at the exact same time.
- Task 4.6: Real-Time Transit Tracking
 - Data Entities: TRACKING_TELEMETRY, VEHICLE, TRIP_ASSIGNMENT
 - Quality Attribute: 7.2 Performance and Scalability (*Latency Bounds*)
 - Validation Defense [C, R]: Keeps ingestion data processing time under 2 seconds for high-frequency GPS tracking streams. This ensures the live dashboard stays fast and accurate without lagging out.
- Task 4.7: Handle Delivery Exceptions
 - Data Entities: DELIVERY_EXCEPTION, TRIP_ASSIGNMENT, VEHICLE
 - Quality Attribute: 7.6 Fault Tolerance and Recoverability (*Dead Letter Queues*)
 - Validation Defense [C, R, U]: Isolates unexpected shipping issues—like truck breakdowns or cargo issues—away from the main pipeline. It dumps them into separate streams so managers can handle them right away without slowing down other shipments.
- Task 4.8: Process Order Modifications
 - Data Entities: ORDER, VEHICLE, INVOICE
 - Quality Attribute: 7.5 Reliability and Data Integrity (*Idempotent API Design*)
 - Validation Defense [R, U]: Implements unique transaction tracking tokens for every update request. If a customer clicks submit multiple times due to a bad network connection, the system only processes the data change exactly once.
- Task 4.9: Automated Invoice Generation
 - Data Entities: INVOICE, ORDER, PAYMENT_TRANSACTION
 - Quality Attribute: 7.5 Reliability and Data Integrity (*ACID Compliance*)
- Task 4.10: Payment Reconciliation
 - Validation Defense [C, R]: Groups basic freight costs, fuel surcharges, and tax calculations into a single database transaction. If any part of the calculation errors out, the entire invoice rolling update is wiped completely to avoid corrupt financial records.
 - Data Entities: PAYMENT_TRANSACTION, INVOICE
 - Quality Attribute: 7.3 Security and Privacy (*System Governance / Audit Trails*)
 - Validation Defense [C, R, U]: Creates locked, unchangeable daily transaction summaries. These are cross-checked automatically against external banking statements to make sure our audit logs are clean and fully secure.

Part I: Conclusion

In evaluating the deployment options for the Smart Fleet Management System (SmartFM), Solution 3 (Hybrid Decentralized Edge Architecture) was omitted from the roadmap. Although this approach theoretically allows offline processing at localized ABC-Trans hubs, it introduces severe architectural drawbacks. Specifically, setting up data replication across multi-master configurations induces high synchronization latency, which directly violates the requirement for real-time tracking. Furthermore, maintaining localized hardware across all branches in Vietnam drastically escalates infrastructure costs, thereby contradicting the core need for centralized logistics visibility. Consequently, the technical roadmap narrows focus onto the two most practical cloud options: Solution 1 (Cloud-Native CQRS Microservices) and Solution 2 (Cloud-Hosted Monolithic Architecture).

Concurrently, this Software Requirements Specification establishes the complete, validated blueprint for SmartFM, bridging the gap between legacy procedures and a high-performance logistics platform. By utilizing the task-driven methodology from Swinsoft Consulting, the framework proves that all 10 core logistics, fleet, and financial workflows map cleanly into structural data entities without implementation lock-in. As a result, backed by rigorous conceptual data models and comprehensive non-functional requirements, the system ensures high scalability, security, and resilience under peak traffic. Ultimately, this specification satisfies all engineering goals, leaving the SmartFM platform fully validated and ready for full-scale cloud deployment.

